

Tarun Kumar

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Summary

Robotics & Automation Engineering student (B.E., expected 2027) specializing in ROS2 autonomous navigation, UAV systems, and computer vision. Team Lead – Autonomous Systems & Path Planning for ISRO's Bharatiya Antariksh Hackathon 2026; CV/Autonomy Lead for a competition UAS in SAEINDIA AeroTHON 2026. Active open-source contributor to ROS2 Navigation2.

Education

Sahyadri College of Engineering and Management

B.E. in Robotics and Automation Engineering

2023 – Present

Mangaluru, India

Experience

Project Intern

Feb 2025 – May 2025

National Institute of Technology Karnataka (NITK)

Mangaluru, India

- Designed and structurally validated a portable catapult launch system for fixed-wing UAVs using SolidWorks and ANSYS Workbench FEA simulation.

Team Member – Robotics Team

Oct 2023 – Feb 2025

Team Challengers

Mangaluru, India

- Built and flight-tested UAV hardware including the AgriDrone Audit System; integrated AI-based crop monitoring with real-time onboard image processing.
- Implemented GPS navigation, flight-controller tuning, and redundant failsafe protocols for autonomous mission execution.

Design Assistant Intern

Jun 2024 – Dec 2024

Atharva Enterprise

Mangaluru, India

- Designed mechanical components and 3D models in SolidWorks; supported design reviews, technical drawings, and iterative prototyping.

Projects

ICE-PATH – Lunar Ice Detection & Rover Traverse Planning | Python, FastAPI, React, Vite

- Leading the autonomous systems & path-planning track for ISRO's Bharatiya Antariksh Hackathon 2026 (Challenge 08): subsurface lunar-ice detection and rover-traverse planning using Chandrayaan-2 DFSAR and OHRC/TMC-2 data.
- Built an in-browser weighted A* path planner with a React/Vite dashboard for landing-site and rover-traverse visualization.

AeroTHON 2026 (SAEINDIA) – Autonomous UAS for Surveillance & Delivery | ROS2, PX4, MAVROS, YOLOv8

- Autonomy and computer-vision lead for a 1.8kg X-frame quadrotor; built the ROS2 Humble + MAVROS OFFBOARD pipeline with a YOLOv8 detection node and PID-corrected precision payload drop.
- Presented the team's Technical Design Report and MATLAB-based stability analysis to an industry/academic review panel at BMS College of Engineering, Bangalore.

Autonomous Mobile Robot | ROS2 Jazzy, Nav2, SLAM Toolbox, Raspberry Pi 5

- Built a custom differential-drive robot on physical hardware with LiDAR-based SLAM mapping and Nav2 path planning for real-time indoor navigation.

Intelligent Autonomous Warehouse Picking Robot | ROS2, OpenCV, ArUco, A*

- Built a camera-based item-detection and dynamic task-rescheduling system for an autonomous picking robot with Team SudoSquad at a robotics/AI hackathon.

Technical Skills

Robotics/Autonomy: ROS2 (Humble, Jazzy), Nav2, SLAM Toolbox, Cartographer, Path Planning, Sensor Fusion

UAV/Aerospace: PX4, ArduPilot, MAVROS, MAVLink, Pixhawk, GPS Navigation, Mission Planner

Programming: Python, C, C++, Bash

Computer Vision/AI: YOLOv8, YOLOv5, OpenCV, TensorFlow, Scikit-learn

Embedded: Jetson Nano, Raspberry Pi, ESP32, Arduino, Intel RealSense

Simulation/CAD: Gazebo, MATLAB/Simulink, ANSYS Workbench, SOLIDWORKS (CSWA), Fusion 360

Leadership, Open Source & Certifications

Open Source – Contributed Costmap2D documentation and parameter-clarity improvements to **ros-navigation/nav2**; pull request merged by maintainer.

Certifications – AI & ML in Space Exploration (India Space Academy) • SOLIDWORKS CAD Design Associate (CSWA) • MATLAB & Simulink Onramp • Python Foundation Certification (Infosys Springboard)